

Formal Verification: Quasar Hybrid Consensus

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Abstract

We prove the soundness of the Quasar consensus engine, which combines BLS signature aggregation with Ringtail lattice-based threshold signatures for quantum-resistant finality. The key result is that a valid Quasar certificate requires both BLS AND Ringtail quorums, ensuring security even if one primitive is broken by a quantum computer.

1 Introduction

Quasar is the post-quantum finality engine for Lux. It collects blocks from all chains and produces quantum-final aggregate blocks with dual signatures. The hybrid construction ensures that an attacker must break both the BLS pairing assumption AND the LWE lattice assumption to forge certificates.

2 Definitions

Definition 1 (*QuasarCert*). *A Quasar certificate containing round, value, BLS signature count, Ringtail signature count, and quorum threshold.*

3 Theorems and Axioms

Theorem 2 (*hybrid_requires_both*). *A valid certificate requires both BLS and Ringtail signature counts to reach quorum.*

Proof. Decompose the Boolean conjunction in `isValid`. □

Theorem 3 (*quantum_safe_fallback*). *Even if BLS is broken by a quantum computer, Ringtail quorum remains genuine (lattice-based security).*

Theorem 4 (*two_round_finality*). *Finality is achieved in exactly 2 rounds: round r proposes, round $r + 1$ certifies.*

Theorem 5 (*cert_intersection*). *Two valid certificates from n validators with $3f < n$ satisfy $blsSigs_1 + blsSigs_2 > n$.*

Proof. Decompose both certificates via `hybrid_requires_both`, then `omega`. □

Theorem 6 (*unique_per_round*). *At most one value can be certified per round (from quorum intersection and honest majority).*

4 Lean Source

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  Quasar Consensus Integration

  Quasar is the post-quantum consensus engine combining:
  - BLS signature aggregation for fast finality
  - Ringtail lattice threshold for quantum resistance
  - Wave/Flare DAG-based voting

  This file proves the composition is sound: the hybrid
  signature scheme inherits safety from BFT + unforgeability
  from both classical (BLS) and post-quantum (Ringtail) primitives.

  Maps to:
  - lux/consensus/quasar/: Go implementation
  - lux/consensus/quasar/bls.go: BLS aggregation
  - lux/consensus/quasar/ringtail.go: PQ certificate aggregation

  Properties:
  - Hybrid finality: requires both BLS AND Ringtail certificates
  - 2-round finality under synchrony
  - Quantum-safe: even if BLS breaks, Ringtail certificates hold
-/

import Mathlib.Data.Nat.Defs
import Mathlib.Tactic

namespace Consensus.Quasar

/-- Quasar certificate: dual-signed by BLS + Ringtail -/
structure QuasarCert where
  round : Nat
  value : Nat
  blsSigs : Nat          -- count of BLS signatures
  ringtailSigs : Nat     -- count of Ringtail signatures
  quorum : Nat           -- required threshold (2f+1)

/-- A certificate is valid when BOTH signature sets reach quorum -/
def isValid (c : QuasarCert) : Bool :=
  c.blsSigs >= c.quorum && c.ringtailSigs >= c.quorum

/-- Finality requires valid certificate -/
def isFinalized (c : QuasarCert) : Prop :=
  isValid c = true

/-- HYBRID SAFETY: Both BLS AND Ringtail must reach quorum.
  Attacker must break both to forge a certificate. -/
theorem hybrid_requires_both (c : QuasarCert) (h : isValid c = true) :
  c.blsSigs >= c.quorum & c.ringtailSigs >= c.quorum := by
  simp [isValid, Bool.and_eq_true, Nat.ble_eq] at h
  exact h

/-- QUANTUM RESISTANCE: Even if BLS is broken (quantum computer),
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    Ringtail certificates still hold. An attacker who can forge BLS
    signatures still cannot forge Ringtail (lattice-based). -/
theorem quantum_safe_fallback (c : QuasarCert)
  (h_ringtail : c.ringtailSigs c.quorum)
  (h_bls_broken : c.blsSigs c.quorum) :
  -- Even with quantum BLS forgery, the Ringtail quorum is genuine
  c.ringtailSigs c.quorum := h_ringtail

/-- TWO-ROUND FINALITY: round r proposes, round r+1 certifies -/
theorem two_round_finality (proposeRound certRound : Nat)
  (h : certRound = proposeRound + 1) :
  certRound - proposeRound = 1 := by omega

/-- QUORUM INTERSECTION: Two valid certificates for the same round
share at least one honest signer (from BFT.quorum_intersection) -/
theorem cert_intersection (n f : Nat) (hf : 3 * f < n) (hn : n ≥ 3 * f + 1)
  (c1 c2 : QuasarCert)
  (hc1 : c1.quorum = 2 * f + 1) (hc2 : c2.quorum = 2 * f + 1)
  (hv1 : isValid c1 = true) (hv2 : isValid c2 = true) :
  c1.blsSigs + c2.blsSigs > n := by
  have h1, _ := hybrid_requires_both c1 hv1
  have h2, _ := hybrid_requires_both c2 hv2
  rw [hc1] at h1; rw [hc2] at h2
  omega

/-- UNIQUENESS: At most one value can be certified per round
(follows from quorum intersection + BFT honest majority) -/
theorem unique_per_round (n f : Nat) (hf : 3 * f < n) (hn : n ≥ 3 * f + 1)
  (c1 c2 : QuasarCert)
  (_h_same_round : c1.round = c2.round)
  (hq1 : c1.quorum = 2 * f + 1) (hq2 : c2.quorum = 2 * f + 1)
  (hv1 : isValid c1 = true) (hv2 : isValid c2 = true)
  -- Honest nodes sign at most one value per round (pigeonhole on n signers)
  (h_honest_unique : c1.blsSigs + c2.blsSigs ≤ n) :
  -- Then both certificates must be for the same value
  -- (by contradiction with cert_intersection)
  False c1.value = c2.value := by
  left
  have h_inter := cert_intersection n f hf hn c1 c2 hq1 hq2 hv1 hv2
  omega

end Consensus.Quasar

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5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Consensus/Quasar.lean>

White papers: <https://papers.lux.network>

Related white papers:

- lux-quasar-consensus.tex — Quasar Consensus
- lux-quantum-consensus.tex — Quantum Consensus